

ZINNWALD GEORGENFE, GermanyWMO: **105820**Lat: **50.7314N**Lon: **13.7517E**Elev: **2877**StdP: **13.23**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.3	7.8	-4.1	4.9	5.4	2.0	6.8	9.6	34.8	32.0	31.8	30.8	10.0	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	73.8	61.4	70.8	60.1	68.0	58.6	63.3	71.4	61.4	68.6	59.7	66.1	10.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.0	86.1	67.2	58.4	81.0	65.2	56.7	76.4	63.2	30.2	71.4	28.8	68.5	27.5	66.2	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.1	26.5	23.7	DB	3.4	80.1	6.8	3.2	-1.5	82.4	-5.5	84.3	-9.3	86.0	-14.2	88.3
			WB	2.5	66.9	6.7	1.6	-2.3	68.1	-6.2	69.0	-10.0	69.9	-14.8	71.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	24.8	24.0	31.1	38.1	47.9	53.1	57.2	58.1	50.0	42.3	32.2	27.9
	DBStd	14.00	8.95	9.11	7.52	7.38	6.94	6.78	6.43	6.02	5.72	7.09	6.66	7.16
	HDD50	4185	782	727	585	361	125	46	11	7	70	253	534	684
	HDD65	8903	1247	1147	1050	806	530	358	251	226	451	704	984	1149
	CDD50	779	0	0	0	5	60	140	235	256	69	14	0	0
	CDD65	23	0	0	0	0	0	2	10	11	0	0	0	0
	CDH74	85	0	0	0	0	0	5	33	47	0	0	0	0
	CDH80	5	0	0	0	0	0	0	2	3	0	0	0	0
	WSAvg	12.0	14.1	13.8	13.3	11.1	10.7	10.3	9.7	9.2	10.8	13.4	13.7	14.2
	PrecAvg	27.3	1.8	1.6	1.9	1.5	2.6	3.1	3.5	3.3	2.2	1.9	2.0	2.0
(15) Precipitation	PrecMax	39.2	3.5	2.9	6.2	3.1	5.2	6.4	6.3	9.0	5.0	3.9	4.2	4.2
	PrecMin	16.2	0.2	0.4	0.6	0.1	0.5	1.4	0.9	0.4	0.8	0.4	0.0	0.6
	PrecStd	4.5	0.9	0.7	1.1	0.7	1.3	1.2	1.3	1.9	1.1	0.9	1.0	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	48.9	54.6	60.7	67.0	74.1	78.1	79.1	68.4	62.9	52.9	49.1
		MCWB	41.2	41.7	46.6	50.0	55.3	61.0	64.8	64.6	58.2	55.8	44.8	43.2
	2%	DB	39.6	43.4	48.0	56.8	64.5	70.0	73.8	74.3	65.4	57.2	47.9	44.3
		MCWB	37.5	39.5	42.7	47.9	53.6	59.4	61.0	62.3	57.1	52.1	43.6	41.0
	5%	DB	37.9	38.3	43.9	52.9	61.9	66.8	71.1	70.6	62.4	54.1	44.5	41.0
		MCWB	36.1	35.6	40.1	45.6	52.3	57.6	60.0	60.6	55.6	50.3	42.2	38.7
	10%	DB	35.5	35.0	40.7	49.3	59.1	63.8	67.8	67.8	59.7	51.6	41.4	38.0
		MCWB	33.6	33.6	38.1	43.2	50.9	56.1	58.7	58.9	54.0	48.6	39.7	36.4
	0.4%	WB	41.3	43.1	47.7	51.2	57.3	63.4	65.5	65.9	60.0	56.6	46.8	44.7
		MCDB	42.8	47.0	51.5	58.2	63.5	69.9	75.8	75.0	65.9	61.8	49.1	48.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.4	39.6	43.0	48.7	55.3	61.3	63.2	63.7	58.3	53.4	44.8	41.7
		MCDB	39.2	43.2	46.9	55.4	62.4	67.1	71.1	72.7	63.8	56.2	47.1	43.9
	5%	WB	36.6	36.3	40.7	46.6	53.3	58.9	61.2	61.5	56.6	51.0	42.4	39.2
		MCDB	37.4	38.0	43.4	51.7	60.3	64.4	68.6	69.3	61.6	53.5	44.5	40.5
	10%	WB	33.9	33.5	38.2	43.8	51.4	56.9	59.3	59.8	54.4	49.0	39.7	36.7
		MCDB	34.9	34.7	40.6	48.6	58.2	62.9	66.1	66.4	59.1	51.2	41.5	37.8
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	7.7	8.4	11.5	12.8	12.3	12.9	12.7	10.8	9.4	7.7	6.7
		MCDBR	10.5	11.6	13.5	16.2	16.0	15.5	16.8	16.4	14.1	12.0	12.6	9.6
		MCWBR	9.2	9.1	9.8	10.4	10.0	9.5	9.8	9.5	9.1	8.4	9.8	7.9
	5% WB	MCDBR	9.3	10.7	12.9	15.3	14.6	14.3	15.3	15.5	13.6	10.9	11.5	9.2
		MCWBR	9.1	8.7	9.9	10.3	10.0	9.4	9.9	9.7	9.0	8.2	9.6	8.2
(40) Clear-Sky Solar Irradiance	taub	0.287	0.309	0.349	0.389	0.390	0.386	0.400	0.395	0.361	0.337	0.312	0.282	(40)
	taud	2.392	2.374	2.305	2.236	2.261	2.294	2.257	2.293	2.369	2.420	2.441	2.411	(41)
	Ebn at Noon	238	262	269	270	274	276	269	265	262	247	226	222	(42)
	Edh at Noon	19	26	33	40	41	40	41	37	31	24	19	16	(43)
(44) All-Sky Solar Radiation	RadAvg	251	493	802	1278	1542	1661	1576	1375	984	577	274	191	(44)
	RadStd	43	76	114	167	184	140	180	131	134	75	35	22	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (56 neighbors)	+0.72	N/A	N/A	N/A	+0.61	+0.61	N/A	N/A	+90	+29

Nomenclature: See separate page